

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
AC Power Supply System, Single Inverter - Automatic Switch-over	105-99034	3.0	4200	12600
AC Power Supply System, Single Inverter - Manual Switch-over	105-990766	2.8	4200	11760
AC Power Supply System, Dual Inverters - Automatic Switch-over	105-991606	6.2	4200	26040

7. SYSTEM DESCRIPTION

EFFECTIVITY *Single-inverter System with automatic switch-over*

The AC power supply system (see Fig. 1) converts 28V direct current from the helicopter electrical system into 26V and 115V alternating current to power various customer defined equipment. The system comprises a main static inverter, a standby static inverter (an integral component of the directional gyro), a control relay and two 26V AC busbars (and third 115V AC busbar, as required). The main inverter receives 28V DC power from the main bus PP21 via an INV circuit breaker when the INV toggle switch is on.

Failure of the main inverter is indicated by illumination of the AC FAIL caution light. The control relay activates the caution light and simultaneously switches AC power from the standby inverter to the 26V AC emergency bus and deactivates the 26V AC normal bus. The standby inverter is powered from the main bus PP21 (or optional emergency bus PP20, FMS 10-36) via the DG circuit breaker when the DG toggle switch is on.

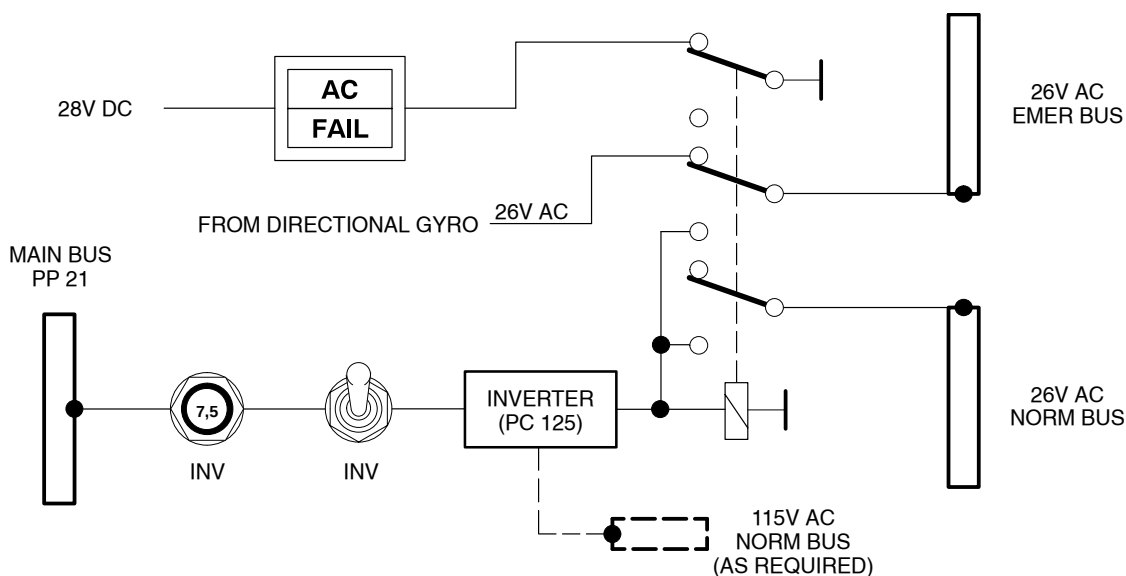


Fig. 1 Typical Single-inverter System with Automatic Switch-over

EFFECTIVITY *Single-inverter System with manual switch-over*

The AC power supply system (see Fig. 2) converts 28V direct current from the helicopter electrical system into 26V and 115V alternating current to power various customer defined equipment. The system comprises a main static inverter, a standby static inverter (an integral component of the directional gyro), an AC POWER toggle switch and two 26V AC busbars (and third 115V AC busbar, as required). The main inverter receives 28V DC power from the main bus PP21 via an INV circuit breaker when the INV toggle switch is on.

Failure of the main inverter is indicated by illumination of the AC FAIL caution light. Placing the guarded AC POWER switch to the EMER position switches AC power from the standby inverter to the 26V AC emergency bus and deactivates the 26V AC normal bus. The standby inverter is powered from the main bus PP21 (or optional emergency bus PP20, FMS 10-36) via the DG circuit breaker when the DG toggle switch is on.

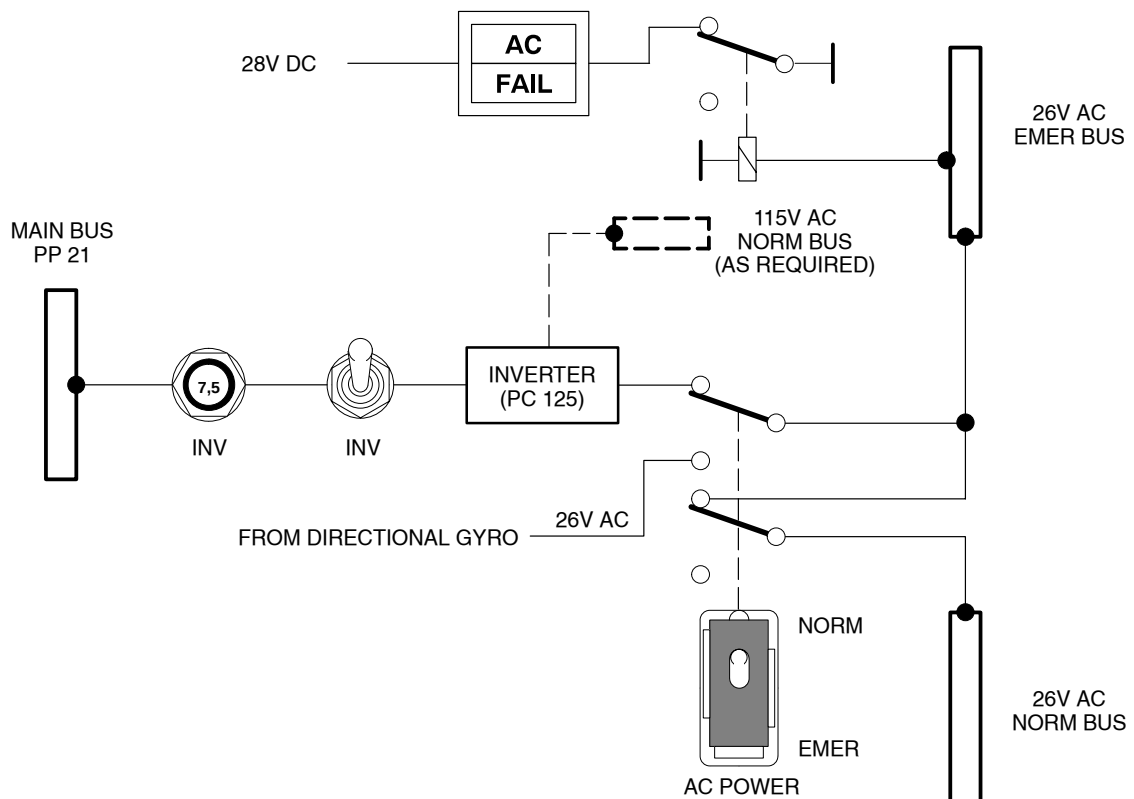


Fig. 2 Typical Single-inverter System with Manual Switch-over

EFFECTIVITY *Dual-inverter System with automatic switch-over*

The AC power supply system(see Fig. 3) converts 28V direct current from the helicopter electrical system into 26V and 115V alternating current to power various customer defined equipment. The system comprises two static inverters, two control relays a 26V AC bus-bar, and a 115V AC busbar. Inverter 1 receives 28V DC power from the emergency bus PP20 via an INV 1 circuit breaker. Inverter 2 receives 28V DC power from the main bus PP21 via an INV 2 circuit breaker. Inverter 1 operates as the primary AC power source with inverter 2 operating in a standby mode.

Failure of an inverter is indicated by illumination of the respective INV 1 or INV 2 caution light. The inverter control relay activates the respective caution light and simultaneously switches the faulty inverter off-line.

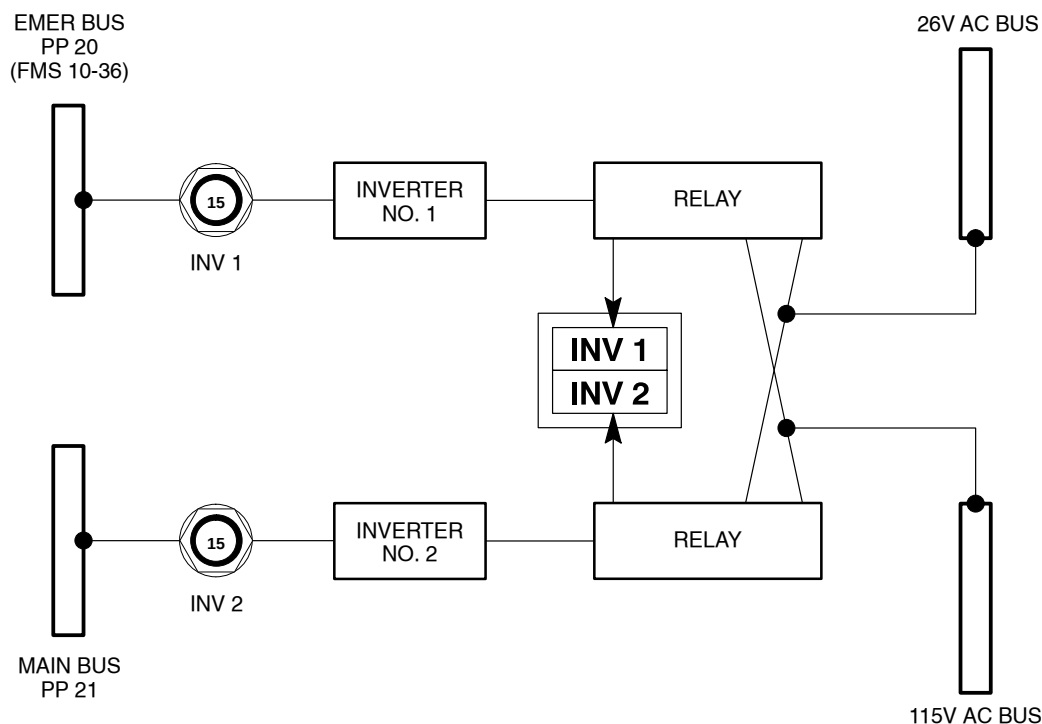


Fig. 3 Typical Dual-inverter System with Automatic Switch-over